



Grade 4 – Term 1 Formal Practical

Living and Non-Living Things

Subject: _____

Learner: _____

School: _____

Date: _____

Mark: _____

Practical Investigation: Yeast – A Living Organism

Aim

To investigate what dry yeast needs in order to become active (“start living” again).

Background Information

Yeast is a living organism. When it is dry, it is inactive (dormant). When it is given the right conditions, it becomes active and produces a gas called carbon dioxide. This process is called **fermentation**.

Apparatus / Materials

- A packet of dry yeast
- Warm water (not boiling)
- Sugar
- 100 ml glass beaker or clear container
- A spoon
- A balloon
- A rubber band



- Test tube
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Method

1. Pour warm water into the glass beaker.
 2. Add one spoon of sugar and stir until it dissolves completely and pour to the test tube.
 3. Add the dry yeast to the sugar solution in the test tube.
 4. Stir gently.
 5. Carefully stretch the balloon over the mouth of the test tube.
 6. Secure the balloon with a rubber band.
 7. Observe what happens every 30 minutes.
 8. Record your observations.
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Safety / Precautions

- Do NOT use boiling water. Very hot water will kill the yeast.
 - Handle glass carefully.
 - Wash your hands after the experiment.
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Observations

Time	Observation
0 min	



10 min	
20 min	
30 min	

Questions

1. What did you observe after the first 10 minutes? (2)

2. Why did you add sugar to the water? (2)

3. What does dry yeast need to become active? (3)

4. What gas filled the balloon? (2)

5. Why must the water be warm and not boiling? (2)

6. What would happen if no sugar was added? (2)

7. What would happen if cold water was used? (2)



8. Which life process can be observed in this activity? Explain your answer. (3)

9. Is yeast a living or non-living thing? Give TWO reasons for your answer. (4)

Conclusion

Write a short paragraph explaining what you learned about what yeast needs to become active. (4)

Total: 20 Marks